THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** cell	Cell.
2	**(c)** water	Through a membrane.
3	**(b)** two identical cells	Growth.
4	**(b)** Robert Hooke	1665.
5	**(b)** high to low concentration	Gradient.
6	**(c)** gametes	Half chromosomes.
7	**(d)** nucleus	Contains DNA.
8	**(d)** swells	Water enters.
9	**(b)** 46	23 pairs.
10	**(d)** cell wall	Cellulose.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Basic structural and functional unit of life (2).	2
12	Cells without a nucleus (1); Bacteria (1).	1+1
13	Photosynthesis (2).	2
14	Schleiden (1); Schwann (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell wall (1); Plastids (1); Vacuole size (1).	1+1+1
16	Protects (1); Controls entry (1); Controls exit (1).	1+1+1
17	Nucleus (1); Mitochondria (1); Chloroplasts (1).	1+1+1
	OR: Nucleus (1); Mitochondria (1); Ribosomes (1).	1+1+1
18	Control centre (1); Contains DNA (1); Heredity (1).	1+1+1
	OR: Made of cells (1); Basic unit (1); From pre-existing cells (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Membrane (1); Nucleus (1); Cytoplasm (1); Organelles (1); Functions (1).	1+1+1+1+1
	OR: Process (1); Two cells (1); Growth (1); Repair (1); Same chromosomes (1).	1+1+1+1+1
20	Wall (1); Membrane (1); Plastids (1); Vacuole (1); Nucleus (1).	1+1+1+1+1
	OR: Process (1); Four cells (1); Half chromosomes (1); Gametes (1); Variation (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bacteria have a nucleus, or that prokaryotes have no genetic material; the DNA is present but is not enclosed in a nuclear membrane.
- Mixing up hypotonic and hypertonic: in a *hypotonic* outside solution the cell swells, in a *hypertonic* one it shrinks.
- Calling osmosis the movement of any solute; osmosis is the movement of *water* through a selectively permeable membrane.
- Confusing the functions of rough ER (protein synthesis, studded with ribosomes) and smooth ER (lipids, detoxification), or saying that mitochondria 'make' energy; they release energy from food as ATP.
- Saying that the cell wall and cell membrane are the same; the membrane is selectively permeable and present in all cells, the wall is rigid and fully permeable.
- Forgetting that plant cells do not burst in water because of the cell wall, whereas an animal cell like RBC can swell and burst.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** shrinks	Water leaves.
2	**(b)** ribosomes	Ribosome subunits.
3	**(c)** mitochondria	Energy.
4	**(b)** swell	Endosmosis.
5	**(d)** DNA	Heredity.
6	**(b)** ribosomes	Synthesis.
7	**(b)** shrinking of cytoplasm	In hypertonic solution.
8	**(b)** plant cells	Plants.
9	**(d)** lysosomes	Digestive.
10	**(c)** prokaryotes	No nucleus.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cells with a true nucleus (1); Plant cells (1).	1+1
12	Lysosome (2).	2
13	Movement of water through a selectively permeable membrane (2).	2
14	Gives shape and protection (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Chloroplast (1); Chromoplast (1); Leucoplast (1).	1+1+1
16	Storage (1); Turgidity (1); Waste storage (1).	1+1+1
17	Transport (1); Protein synthesis (rough) (1); Lipid synthesis (smooth) (1).	1+1+1
	OR: Amoeba (1); Paramecium (1); Bacteria (1).	1+1+1
18	Number of cells (1); Chromosome number (1); Purpose (1).	1+1+1
	OR: Function (1); Examples (1); Structure fits function (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diffusion (1); Osmosis (1); Examples (1); Differences (1); Importance (1).	1+1+1+1+1
	OR: Hollow potato (1); Sugar solution (1); Water (1); Level rises (1); Osmosis (1).	1+1+1+1+1
20	Statement (1); Schleiden (1); Schwann (1); Virchow (1); Importance (1).	1+1+1+1+1
	OR: Raisins in water (1); Raisins in sugar solution (1); Swells and shrinks (1); Osmosis (1); Conclusion (1).	1+1+1+1+1

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- Calling osmosis the movement of any solute; osmosis is the movement of *water* through a selectively permeable membrane.
- Confusing the functions of rough ER (protein synthesis, studded with ribosomes) and smooth ER (lipids, detoxification), or saying that mitochondria 'make' energy; they release energy from food as ATP.
- Saying that the cell wall and cell membrane are the same; the membrane is selectively permeable and present in all cells, the wall is rigid and fully permeable.
- Forgetting that plant cells do not burst in water because of the cell wall, whereas an animal cell like RBC can swell and burst.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** colour	Pigments.
2	**(c)** chloroplasts	Chlorophyll.
3	**(b)** cell walls	Cellulose.
4	**(c)** starch	Colourless.
5	**(d)** Amoeba	One cell.
6	**(a)** plant cells	Storage.
7	**(d)** packages materials	Dispatch.
8	**(a)** selectively permeable	Controls passage.
9	**(a)** nucleus	DNA.
10	**(c)** a network of membranes	Transport.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Division giving two identical daughter cells (2).	2
12	Movement of particles from higher to lower concentration (2).	2
13	Protein synthesis (2).	2
14	Division giving gametes with half the chromosomes (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Raisins in water (1); Roots absorbing water (1); Wilting (1).	1+1+1
16	Swells (1); May burst (1); Haemolysis (1).	1+1+1
17	Nuclear membrane (1); Nucleolus (1); Chromatin (1).	1+1+1
	OR: Smell of food (1); Ink in water (1); Gas exchange (1).	1+1+1
18	Water leaves (1); Cytoplasm shrinks (1); Plasmolysis (1).	1+1+1
	OR: Everything made of cells (1); Cells perform functions (1); Cells divide (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Nucleus (1); Size (1); Organelles (1); Examples (1); Complexity (1).	1+1+1+1+1
	OR: Hypertonic solution (1); Water leaves (1); Cytoplasm shrinks (1); Wall remains (1); Example (1).	1+1+1+1+1
20	Nucleus (1); Mitochondria (1); Ribosomes (1); Golgi (1); Lysosomes (1).	1+1+1+1+1
	OR: Shape (1); Protection (1); Prevents bursting (1); Only in plants (1); Cellulose (1).	1+1+1+1+1

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